01 / PURPOSE AND KIT

Record a usable pilot

A practical camera + two-IMU capture procedure for the right arm. Start with one short rehearsal; collect the study takes only after it passes.

1 PILOT

about 20 seconds

12 USABLE TAKES

6 clear + 6 briefly occluded

Six development takes first. Recalibrate, then capture six held-out test takes in a separate block. This is a small pilot design, not a statistical power calculation or an established training dataset.

1

Prepare the capture space

Phone on a fixed support, existing ESP32 rig, two selected sensors, secure straps, laptop, small object, target container and a clear table.

2

Check and calibrate the right arm

Verify sensor identity, allow startup bias capture, align both boards on the table, then capture mounted neutral with the arm straight down.

3

Record both systems and keep the evidence

Start the phone and IMU recording separately. Include visible motion pulses before and after the task. Save original video and session JSON under one take ID, including imperfect or failed takes.

START THE LIVE CHECKER FROM THE REPOSITORY ROOT

bash arm_checker/run.sh --port auto --protocol am

Then open http://127.0.0.1:8765/ . Phone recording is separate.

Two recordings, not automatic synchronization

The checker does not record camera video, synchronize clocks or fuse camera and IMU data. Later manual alignment is a separate analysis step.

What this can support: checking feasibility, sensor identity, continuity and failure cases. What it cannot yet establish: accurate 3D hand position, grasp timing, robot control, or a benefit from sensor fusion.

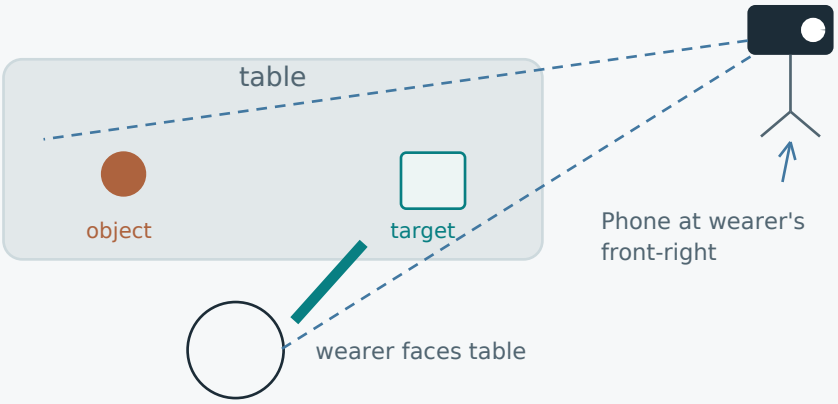
02 / CAMERA PLACEMENT

Keep the whole task in view

Use landscape 1080p at 30 or 60 fps. Keep one frame rate for all takes. A fixed diagonal view from the wearer's front-right is a practical starting point.

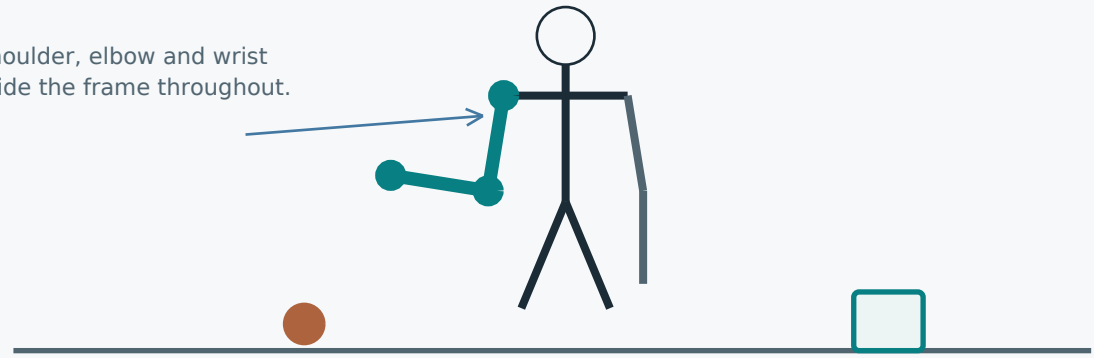
TOP VIEW

Start around 1.5-2 m away. Adjust for complete framing; distance is not a guarantee.



PHONE FRAME / FRONT VIEW

Right shoulder, elbow and wrist stay inside the frame throughout.



Include object, target and the entire hand path. Leave margin.

Before pressing Record: test the full reach, avoid backlighting, keep exposure/focus stable if your phone permits, and make the hand large enough to inspect. Keep the phone position, zoom and table layout unchanged within each capture block.

Raw video versus the current tracker

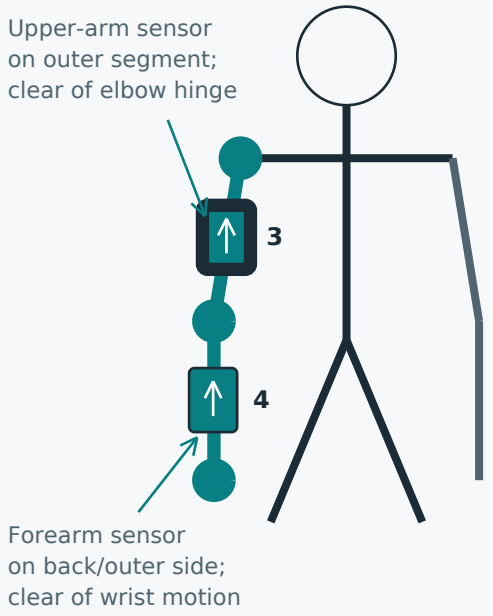
Old visual markers are not required to save a raw pilot video. The current PhoneToPanda vision tracker does require markers; a marker-free replacement is not implemented.

03 / WEARABLE SETUP

Identify first, then mount

Use the right upper-arm and right forearm records from the existing rig. Secure mounts are part of the measurement, not a cosmetic detail.

FRONT VIEW - ANATOMICAL RIGHT



Default AM IDs
3 = right upper arm
4 = right forearm

Move one loose board at a time and watch its live orientation or gyro readout. Confirm the physical board-to-record mapping before putting it on the arm.

Use a repeatable location and record it. Tighten enough to prevent sliding, while remaining comfortable. Do not put a board across a joint. Leave cable slack through the full reach.

Keep all existing wiring and firmware unchanged. This checker selects two records from the five-record stream. No flashing or rewiring is part of this procedure.

Measure the model lengths. Use approximate shoulder-to-elbow and elbow-to-wrist joint-centre distances, in centimetres. Enter those numbers in the checker's cm fields (for example, 30 cm and 26 cm) and on the session sheet. They set the displayed segment lengths; the sensors do not measure them.

Fixed shoulder: keep the torso still

The viewer anchors the shoulder at one point. It does not measure torso translation, chest-relative shoulder angles, independent hand motion or finger closure.

The earlier rig's reported inward/outward reversal has not been physically diagnosed. Verify IDs and movement direction; do not assume a new viewer has fixed it.

04 / CALIBRATION

Three holds before a take

Use Table alignment, then Neutral pose in the viewer. A healthy stream and successful calibration are prerequisites, not proof of physical accuracy.

STARTUP BIAS	TABLE ALIGN	MOUNTED NEUTRAL
10 s	2 s	3 s
Boards stationary	Flat and parallel	Right arm straight down

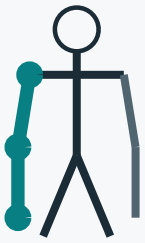
The table and neutral captures also need at least 20 distinct fresh paired samples. Motion or invalid data can restart or prevent capture; wait for the actual completion state.

TABLE ALIGN



Chip/+Z up. Both +X axes toward the same intended forward direction. Keep power on.

MOUNTED NEUTRAL / N-POSE



Arm down; elbow straight; torso still; palm toward thigh.

1 Wait for firmware bias and fresh data

Existing firmware needs ten continuous quiet seconds after startup. Check both selected IDs and valid/fresh indicators.

2 Align on the table, then mount and capture neutral

The table capture aligns relative heading. Mounted neutral assigns sensor mounting offsets so both arm segments point down (-Z).

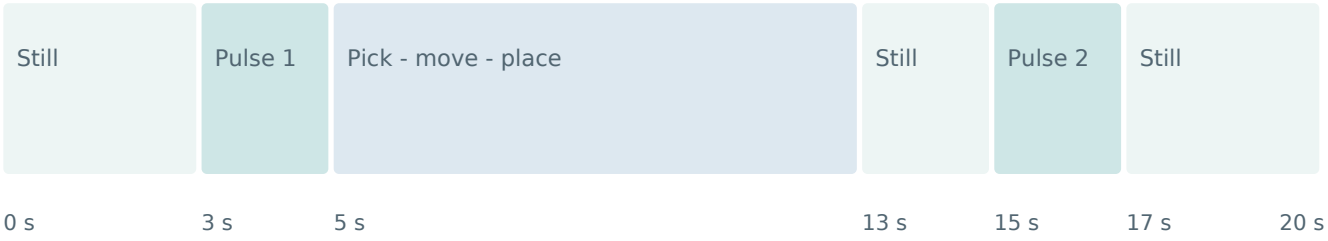
Repeat both captures when the reference changes

Recalibrate after remounting, a device restart, changed sensor IDs or a changed input source. For the separate test block, recalibrate before capturing its first take.

05 / ONE TAKE

A repeatable 20-second sequence

Start phone video first, then start IMU recording. Allow a little video lead-in. Stop IMU recording after the end hold, then stop the phone.



Pulse: with the upper arm fairly still, make one clear, comfortable elbow bend and return. Keep the forearm visible. Use the same gesture before and after the task; avoid tiny movements that are hard to identify in a 30 fps video.

Task: pick up one small object, move it along the same clear path and place it in the target. Aim for about eight seconds; do not rush to fit an exact duration. For occluded takes, hide the camera view for about two seconds during the carry only (page 7).

SEPARATE TIMELINES



Later manual alignment: identify the same visible bend and its IMU angular-motion feature. Estimate an offset using the first pulse; compare the second pulse to check residual timing disagreement. Keep both original files and write down the chosen frames/times and uncertainty.

These gestures provide rough correspondence, not measured calibrated synchronization. The phone, ESP32 and host have independent clocks. The checker records their available timestamps separately; it does not align camera frames automatically.

06 / CAPTURE PLAN

Balance conditions, reserve a block

Use the same wearer, object, target and basic movement for the first pilot. Keep clear and occluded takes balanced within each block.

First: one 20-second pilot, labelled P00

Inspect video framing, both IDs, calibration, session playback and both visible pulses. If anything fails, correct it and repeat the pilot before starting development takes.

Order	Take IDs	Conditions	Use
1	D01 / D02	Clear / Occluded	Development
2	D03 / D04	Occluded / Clear	Development
3	D05 / D06	Clear / Occluded	Development
Break	Recalibrate	New capture block	Freeze analysis choices
4	T01 / T02	Occluded / Clear	Held-out test
5	T03 / T04	Clear / Occluded	Held-out test
6	T05 / T06	Occluded / Clear	Held-out test

Target: 12 usable paired takes. Six development takes (3 clear, 3 occluded), then six test takes (3 clear, 3 occluded). Conditions are specified before capture. A take is a paired video and IMU session, not a single frame.

Keep the test block separate. Recalibrate before it and record that block identity. Choose processing, alignment rules and usability criteria using development data. Do not tune the method against test outcomes and still call those outcomes unseen.

Retain failures and replacements

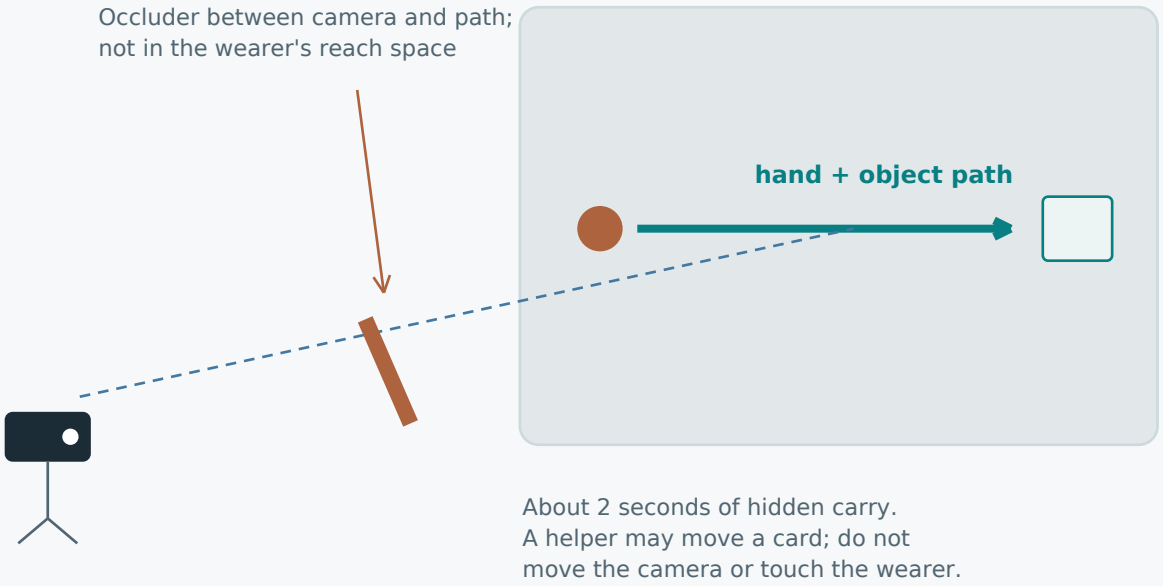
Keep every attempt with a rejection reason. A corrupt or missing recording may be replaced under the same condition; append an attempt suffix such as D02_a02. Do not discard a valid take because the method fails.

07 / OCCLUSION CONDITION

Hide the view, keep the path clear

Create a short, repeatable camera occlusion during carry. Keep start/end poses, timing pulses, pickup and placement visible.

TOP VIEW / ONE CAMERA



Use a card or screen outside the hand path, or ask a helper to cover that line of sight briefly. Do not ask the wearer to reach behind an obstacle. Keep cables slack and the object securely supported. Use the same planned timing across occluded takes.

Check that the intended arm/hand area actually disappears from the camera for the chosen interval. Write down its approximate start and end. Leave the timing pulses and task endpoints unobstructed.

Occlusion is not hidden ground truth

One camera cannot reveal hidden motion. For later error measurement, use an optional second, unoccluded reference camera and verify its timing and geometry.

Continuity during occlusion can be inspected qualitatively. Numerical accuracy claims require an independent reference and a defined evaluation method; the animated arm is not that reference.

08 / FILES AND RECORD SHEET

Make each pair traceable

Keep original phone files and exported session JSON together under a single take ID. Do not overwrite originals after trimming or alignment.

EXAMPLE DIRECTORY

capture_20261011 / development / D01_a01 /

video_original.mp4 imu_session.json notes.txt

Use test / T01_a01 / for the reserved block. Preserve the original phone filename in notes, even if you also create a descriptive copy.

Record for each take	Value / space for notes
Take ID / block / condition	_____
Original video + session ID	_____
Date / wearer / mounting notes	_____
IDs / upper and forearm lengths (cm)	_____
Camera fps / frame size / position	_____
Calibration completed / issues	_____
Pulse 1 / pulse 2 visible?	_____
Occlusion interval, if used	_____
Drops / stale flags / task outcome	_____
Usable? / reason / replacement ID	_____

Before moving on: replay or inspect the saved IMU session, open the original phone clip, confirm both gestures and the full task are present, and check that neither selected sensor became invalid. Save notes while the take is still easy to remember.

Do not pair an old video with a newly recorded IMU movement. A valid pair comes from the same physical event. Exported host and ESP32 timestamps retain their original clock domains.

09 / REVIEW BEFORE ANALYSIS

Pass the checks, keep the limits

Use this page after the rehearsal and after each capture block. A plausible-looking skeleton alone is not sufficient evidence.

If you see this	Check / record this
No stream or wrong device	Close other serial readers; confirm the selected port and protocol. Do not reflash as a first step.
Wrong limb direction	Verify each physical sensor ID separately, then repeat table and mounted calibration. Preserve a mismatch recording.
Capture will not complete	Hold still; check fresh paired samples, sensor validity, gyro motion and mount stability.
Pose freezes or jumps	Inspect stale/invalid states, device restarts and cable movement. Record the fault; do not label held geometry as fresh motion.
Video or pulse is missing	Keep the attempt and note the reason; repeat the same planned condition with a new attempt suffix.
Drift during a take	Keep the evidence. Inspect neutral return and pulse disagreement; do not silently correct the test result.

Measurement boundary. Two six-axis IMUs do not observe absolute heading, global translation, finger closure or independent wrist/hand pose. The fixed-shoulder geometry depends on calibration and measured segment lengths. There is no Panda inverse-kinematics or robot-control connection in this checker.

What is still unverified. Mounted direction correctness, physical angle accuracy, stationary drift and end-to-end latency need measured tests. The earlier inward/outward symptom remains unresolved. This guide proposes data collection; it reports no new physical validation or camera-IMU fusion result.

REVIEWER START - FROM REPOSITORY ROOT

bash arm_checker/run.sh --demo

Open http://127.0.0.1:8765/ . See arm_checker/README.md and hardware_setup.md for live serial input, calibration and the adapter contract. Demo input is synthetic.